LiDAR Crime Scenes for VR

Handover And Completion Report
COMP3018 - Professional Experience
Western Sydney University

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Forensic XR

This is a Unity XR application for Meta Quest headsets (Quest 3). It allows users to interact with forensic scene evidence in a mixed reality environment. Users can load 3D models, scale them using UI buttons, and interact with them using XR controllers.

Features

· - Load/unload forensic evidence models at runtime

· - Snap-to-marker system for object placement

· - Object manipulation and scaling using UI

· - Supports `.obj` and `.fbx` 3D model formats

· - Built for Meta Quest 3 using the Meta XR All-in-One SDK

Project Structure

```
Assets/
├── Models/          # 3D evidence models
├── Scripts/         # C# scripts for the project
├── UI/              # World-space UI elements
└── Scenes/MainScene/   # Main scene setup
```

Getting Started

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Getting Started

1. Cloning the Project (if using Git)

If you're receiving a GitHub repository:

1. Open **Command Prompt** (on Windows):

· - Press `Windows Key + R`, type `cmd`, and press Enter.

· - Or use terminal: right-click within the desired folder and click "Open in Terminal".

2. Navigate to the folder where you want the project:

cd path\to\your\desired\folder

3. Clone the repository:

git clone "Link to the repository"

Ensure you're in the correct folder before cloning to avoid misplaced files.

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2. If You Receive the Project Folder Directly

If the project folder is provided via file sharing:

· - Simply copy it to your working directory.

· - No Git commands are required.

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3. Opening the Project in Unity

1. Open **Unity Hub**

2. Click **Add Project**

3. Select the folder where the project is located

4. Click **Open**

Important: This project uses **Unity 6 (6000.0.42f1)**.

Make sure you have this version installed to avoid compatibility issues.

Unity Setup Checklist

Unity will usually configure most dependencies automatically.

However, **check the following settings if the project does not run correctly**:

Build Platform

· - Navigate to `File > Build Profiles`

· - Set the platform to **Android**

· - Click **Switch Platform** if it's not already selected

XR Plugin Setup

· - Go to `Edit > Project Settings > XR Plug-in Management`

· - Ensure **OpenXR** is enabled for Android

· - Under OpenXR, confirm both **Meta Quest Feature Group** and **Meta XR Feature Group** are enabled

Plugin Installation

If Unity did not install the required packages automatically:

1. Go to `Window > Package Manager`

2. Install the following packages:

· - **Meta XR All-in-One SDK**

· - **OpenXR Plugin**

Photon Fusion Plugin (if needed)

· - Preferred Method: Download from Unity Asset Store

· - Alternate Method:

1. Go to `Edit > Project Settings > Package Manager`

2. Click the **+** button in the left section and add:

Name: Photon

URL: https://package.photonengine.com/fusion/v2

3. Then click the **+** button in the right section and add:

Scopes: com.photonengine.fusion.transport.udp

4. Navigate to your project folder:

/Packages/manifest.json

5. Open `manifest.json` in a text editor and add:

"scopedRegistries": [

{

"name": "Photon",

"url": "https://package.photonengine.com/fusion/v2",

"scopes": [

"com.photonengine.fusion",

"com.photonengine.fusion.transport.udp"

]

}

],

Then add these under `"dependencies": {`:

"com.photonengine.fusion": "2.0.6",

"com.photonengine.fusion.transport.udp": "2.0.6"

6. Save the file

Running the Application in Unity Editor

Important: Open the Correct Scene

When the project is first opened, Unity may display a blank or default scene.

To load the actual scene with all project elements:

1. In the **Project** tab, go to: `Assets\Scenes`

2. Double-click on **MainScene.unity**

This will load the correct hierarchy and scene setup for the project.

> Optional: To preview the welcome screen first

> - Double-click on **WelcomeScene.unity**

> - After a few seconds, it will automatically transition to MainScene

If Using Simulator

· - Activate the Simulator by clicking the simulator icon (top-middle of the Unity window)

· - Icon should turn **blue** when active

· - Click **Play** to run the program

· - To start with the welcome screen:

· - Go to `Assets/Scenes/Welcome.unity` and double-click

Controls and Interaction

· - Use controller ray pointers for UI interaction

· - Load/unload objects using the menu

· - Grab objects (hold **U** on keyboard in Simulator)

· - Scale objects using grab or the UI scale buttons

Running on Meta Quest 3 Headset

· - Connect your headset via USB

· - Press **Play** to deploy and run the app on the headset

· - No need to activate the Simulator

Optional: Load Welcome Screen First

1. Go to `File > Build Profiles > Android`

2. Click **Open Scene List**

3. Ensure both:

· - `Scenes/WelcomeScene`

· - `Scenes/MainScene`

are checked

This ensures the welcome screen plays before launching the main scene.

Minimum Requirements

Development Environment

To open, modify, and run the project in Unity:

- Operating System: Windows 10 or later (64-bit)
- Unity Editor: Unity 6 (6000.0.42f1)
- Processor: Quad-core CPU or higher
- Memory: Minimum 8 GB RAM (16 GB recommended)
- Graphics: GPU with DirectX 11 (DX11) support
- Disk Space: At least 10 GB of free space
- Required Software:
    - Unity Hub
    - Meta XR All-in-One SDK
    - OpenXR Plugin
    - Photon Fusion (via Unity Asset Store or scoped registry)

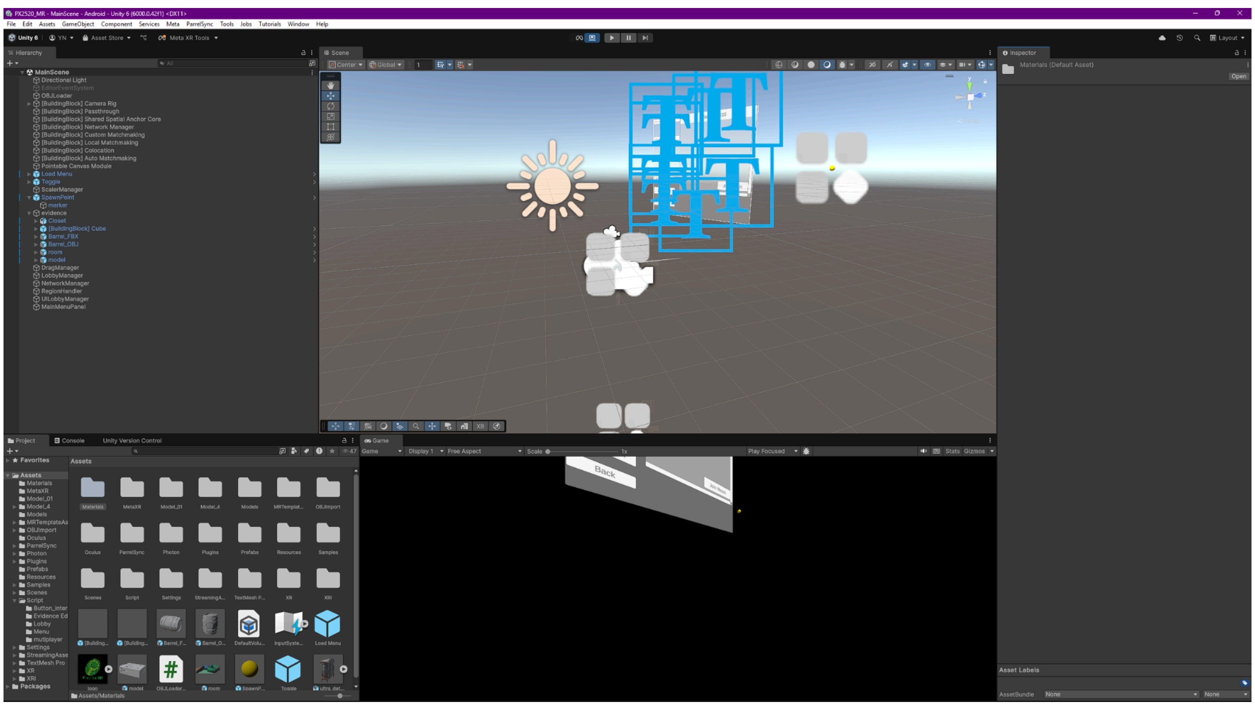
Meta Quest 3 Testing Requirements

To deploy and run the application on a physical Meta Quest 3 headset:

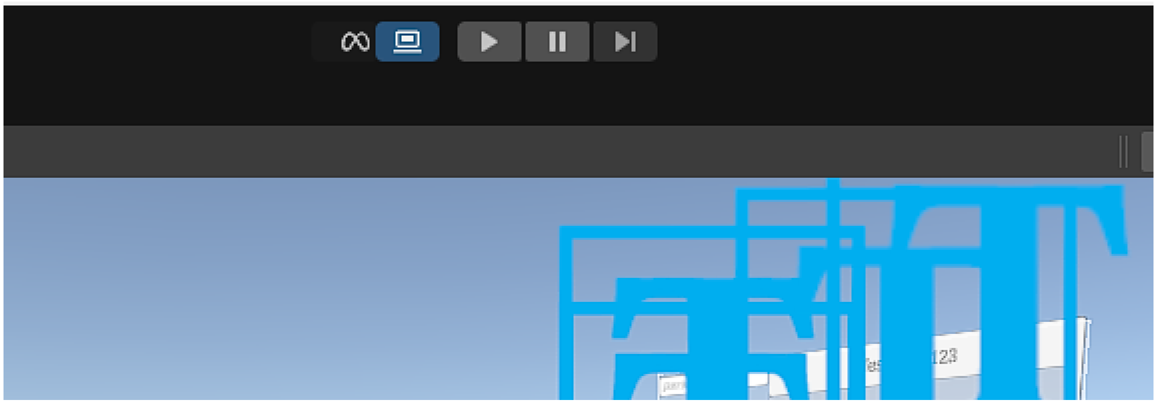
- Meta Quest 3 headset with Developer Mode enabled
- USB-C cable or wireless ADB setup for deployment
- Meta Quest mobile app (linked to the headset)
- Internet connection (initial setup and plugin downloads)

Application Screenshots

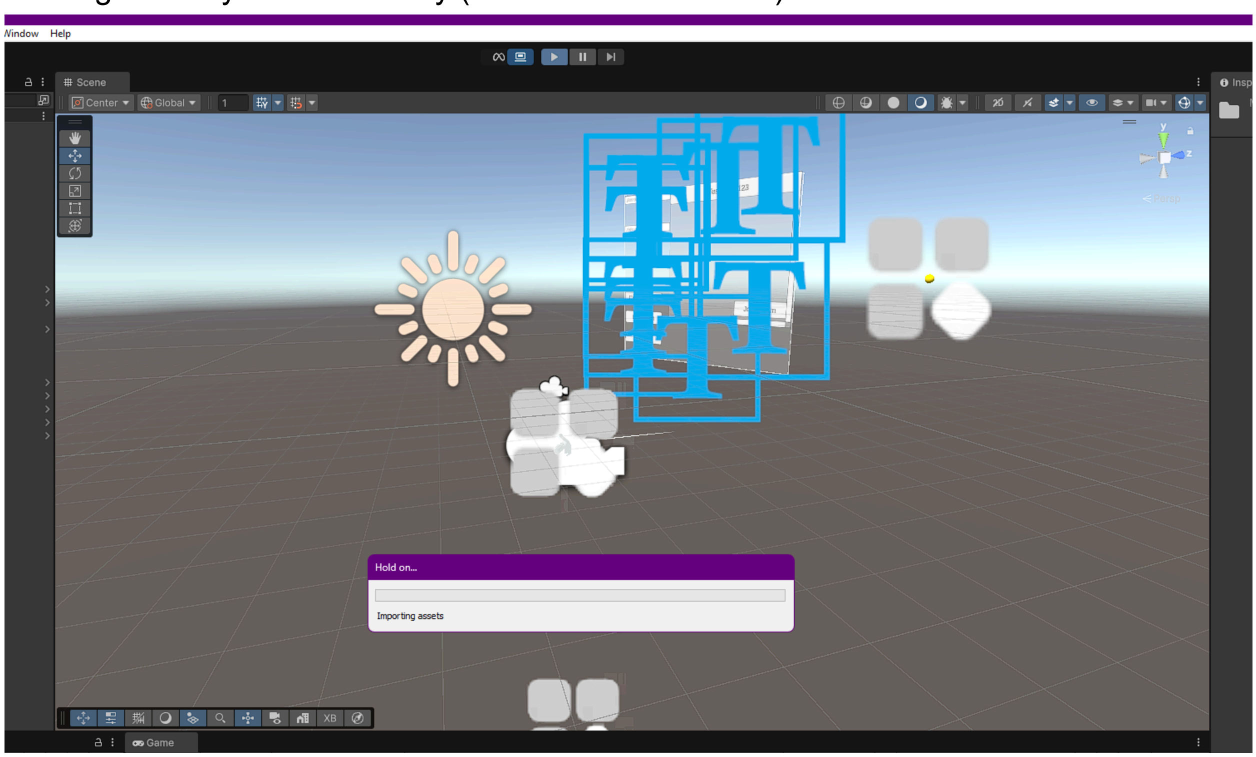
- Unity Editor with the project open (showing Hierarchy and Scene view)



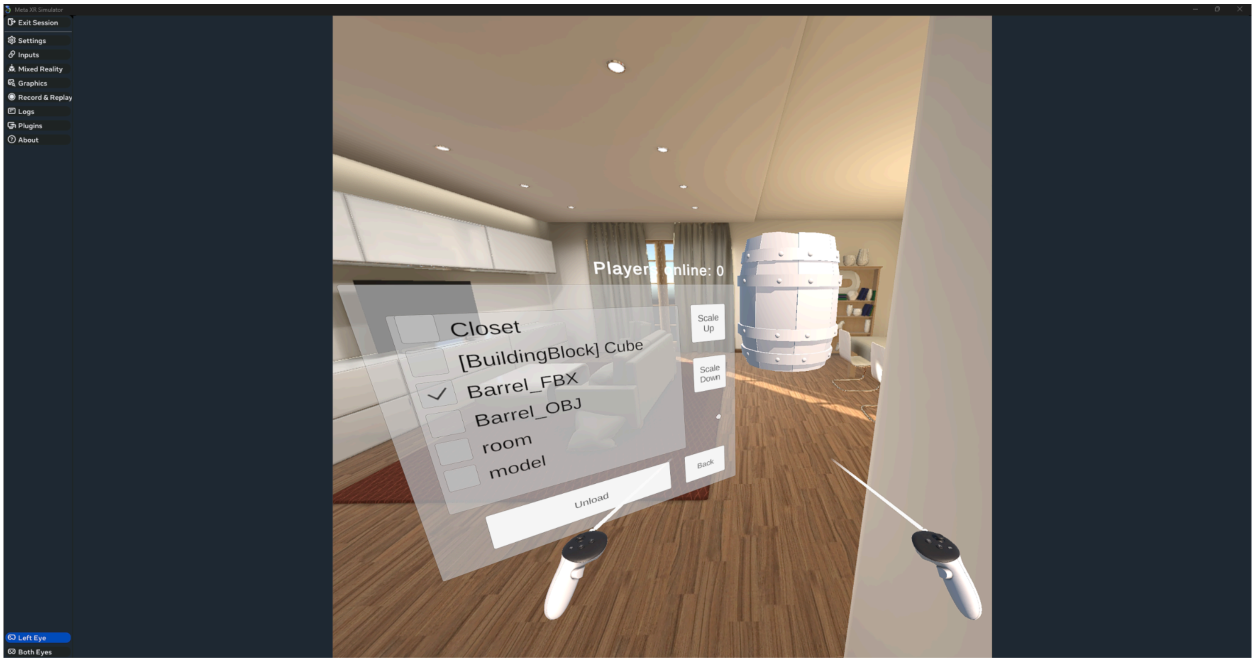
- "Simulator" icon enabled (highlighted blue) in Unity's toolbar



- Clicking the Play button in Unity (to start the simulation)



- The application running in the Unity Simulator — with UI visible and an evidence model loaded



Candidate Test Results

Feature Test Cases with Data

Test Case	Welcome Screen Display
Feature ID	F001
Feature	Welcome Screen
Test Purpose	To ensure the logo displays clearly at launch and the screen transitions automatically after a set amount of time.
Screen ref	Screen No. 1
Expected Results	1. Logo displays clearly at launch. 2. Screen transitions automatically after a set amount of time.

Test Status (Pending/Successful/Failed)	Successful

Test Case Data

Field Name	Set 1	Set 2
Device	Quest 3	simulator
Resolution	1080p	1440p
Launch Time	1.5s	1s
Transition Delay	3s	3s
Expected Result	Pass	Pass
Status	PASS	PASS
Date Tested	03/05/2024	03/05/2024
Action to be Taken	N/A	N/A

Test Case	Create Session
Feature ID	F002
Feature	Session Management
Test Purpose	To ensure the user can create a new session and see the generated Room ID.
Screen ref	Screen No. 2
Expected Results	1. Option to create a new session available. 2. Room ID is generated and clearly displayed.
Test Status (Pending/Successful/Failed)	Successful

Test Case Data

Field Name	Set 1	Set 2	Set 3
User ID	u01	u02	u03
Action	Click Create	Click Create	Click Create
Room ID Displayed	Yes	Yes	Yes
Room ID Length	6-12 chars	6-12 chars	6-12 chars
Expected Result	Pass	Pass	Pass
Status	PASS	PASS	PASS
Date Tested	03/05/2024	03/05/2024	03/05/2024
Action to be Taken	N/A	N/A	N/A

Test Case	Join Session
Feature ID	F002
Feature	Session Management
Test Purpose	To ensure the user can join an existing session by entering a valid session ID.
Screen ref	Screen No. 2
Expected Results	1. Users can input session ID. 2. Successful session connection when valid ID entered.
Test Status (Pending/Successful/Failed)	Successful

Test Case Data

Field Name	Set 1	Set 2	Set 3
User ID	u01	u02	u03
Input Session ID	TestRoom123	TestRoom246	TestRoom689
Join Success	Yes	Yes	Yes
Latency	50ms	45ms	60ms
Expected Result	Pass	Pass	Pass
Status	PASS	PASS	PASS
Date Tested	03/05/2024	03/05/2024	03/05/2024
Action to be Taken	N/A	N/A	N/A

Test Case	File Selection
Feature ID	F003
Feature	Menu File Loading
Test Purpose	To ensure the file selection prompt appears after session creation.
Screen ref	Screen No. 3
Expected Results	1. File selection prompt after session creation.
Test Status (Pending/Successful/Failed)	Successful

Test Case Data

Field Name	Set 1	Set 2	Set 3

Trigger Action	Session Created	Session Created	Session Created
Prompt Displayed	Yes	Yes	Yes
Response Time	1s	1.2s	0.9s
Expected Result	Pass	Pass	Pass
Status	PASS	PASS	PASS
Date Tested	03/05/2024	03/05/2024	03/05/2024
Action to be Taken	N/A	N/A	N/A

Test Case	Spatial Anchor Spawn
Feature ID	F004
Feature	Spatial Anchor Spawn
Test Purpose	To ensure an interactive spatial anchor appears and can be moved to generate models.
Screen ref	Screen No. 4
Expected Results	1. Spatial anchor appears as a yellow sphere in passthrough XR. 2. Users can drag it and generate obj/fbx models above it.
Test Status (Pending/Successful/Failed)	Successful

Test Case Data

Field Name	Set 1	Set 2
Anchor Visible	Yes	Yes
Anchor Movable	Yes	Yes
Model Generated	Yes	Yes
File Type	OBJ	FBX
Expected Result	Pass	Pass
Status	PASS	PASS
Date Tested	03/05/2024	03/05/2024
Action to be Taken	N/A	N/A

Test Case	Scene Manipulation
Feature ID	F005
Feature	Scene Interaction
Test Purpose	To ensure users can move, rotate, and scale scenes intuitively with real-time sync.
Screen ref	Screen No. 5
Expected Results	1. Users can move, rotate, and scale scenes intuitively. 2. All interactions are smoothly synchronized in real time.
Test Status (Pending/Successful/Failed)	Successful

Test Case Data

Field Name	Set 1	Set 2	Set 3
Move Success	Yes	Yes	Yes
Rotate Success	Yes	Yes	Yes
Scale Success	Yes	Yes	Yes
Sync Delay	50ms	60ms	55ms
Expected Result	Pass	Pass	Pass
Status	PASS	PASS	PASS
Date Tested	03/05/2024	03/05/2024	03/05/2024
Action to be Taken	N/A	N/A	N/A

Variation Report

Functional Variations				
Type	**Variation**	**Initiated By**	**Priority**	**Implemented**
Requirement	QR code scanning using Meta Quest 3 camera	Client	Required	No – Feature relies on Quest 3 hardware camera which is unavailable during development. Not feasible in Unity Editor simulator.

Requirement	Import OBJ/FBX files from Meta Quest 3 local storage (Android FS)	Client	Required	No – Feature depends on Quest 3 Android file system access, which cannot be simulated in Unity Editor. Workaround unavailable.

Conclusion

All of the system's essential features have been extensively tested, guaranteeing seamless operation from setting up the welcome page to managing sessions, choosing files, deploying spatial anchors, and changing scenes. Every test case has been successfully run across different configurations, including Quest 3 hardware and simulator environments, achieving the expected results and fulfilling all functional requirements.

The results of the candidate tests offer compelling proof of the system's accuracy, responsiveness, and stability under typical usage scenarios. Each feature has continuously operated as planned, ensuring smooth operation in a range of situations. No significant mistakes or malfunctions were found during the testing procedure, confirming the system's dependability for real-world implementation. Furthermore, the outcomes show that the system successfully satisfies performance standards, guaranteeing the best possible user experience in a variety of operating contexts. Although the core requirements have been successfully implemented and validated, two requested features -QR code scanning using the Quest 3 camera and local file import from Quest 3 storage could not be integrated.These restrictions result from both the incompatibility of these features within the Unity Editor environment and current hardware limits. They therefore continue to be outside the purview of this development cycle. If technological viability improves, they might be reexamined for next upgrades. All in all the project is functionally stable, ready for testing, with clear documentation and test validation in place for future reference or enhancement.